

STAT 408 Lab 07: Logicals

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PUBLISHED
February 26, 2026

Read in Data

```
library(tidyverse)
```

```
Bozeman <- read.csv("Living in Bozeman.csv")
names(Bozeman)
```

```
[1] "Timestamp"
[2] "How.long.have.you.lived.in.Bozeman."
[3] "What.is.something.you.like.to.do.for.fun.in.the.Bozeman.area."
[4] "What.part.of.Bozeman.do.you.live."
[5] "What.is.your.favorite.thing.about.Bozeman..Why."
[6] "What.is.your.favorite.season.in.Bozeman."
[7] "What.are.your.favorite.and.least.favorite.restaurants.in.Bozeman."
[8] "How.far.do.you.travel.to.get.to.campus..miles.."
[9] "How.many.credits.are.you.taking.at.MSU."
```

```
names(Bozeman) = c("timestamp", "years_live_boz", "for_fun_boz", "part_boz_live", "fav_thi
```

Explore the Data

- The variables with a limited set of fixed possible areas are: `years_live_boz`, `fav_season_boz`. This is because these are both the multiple choice questions.

```
table(Bozeman$years_live_boz)
```

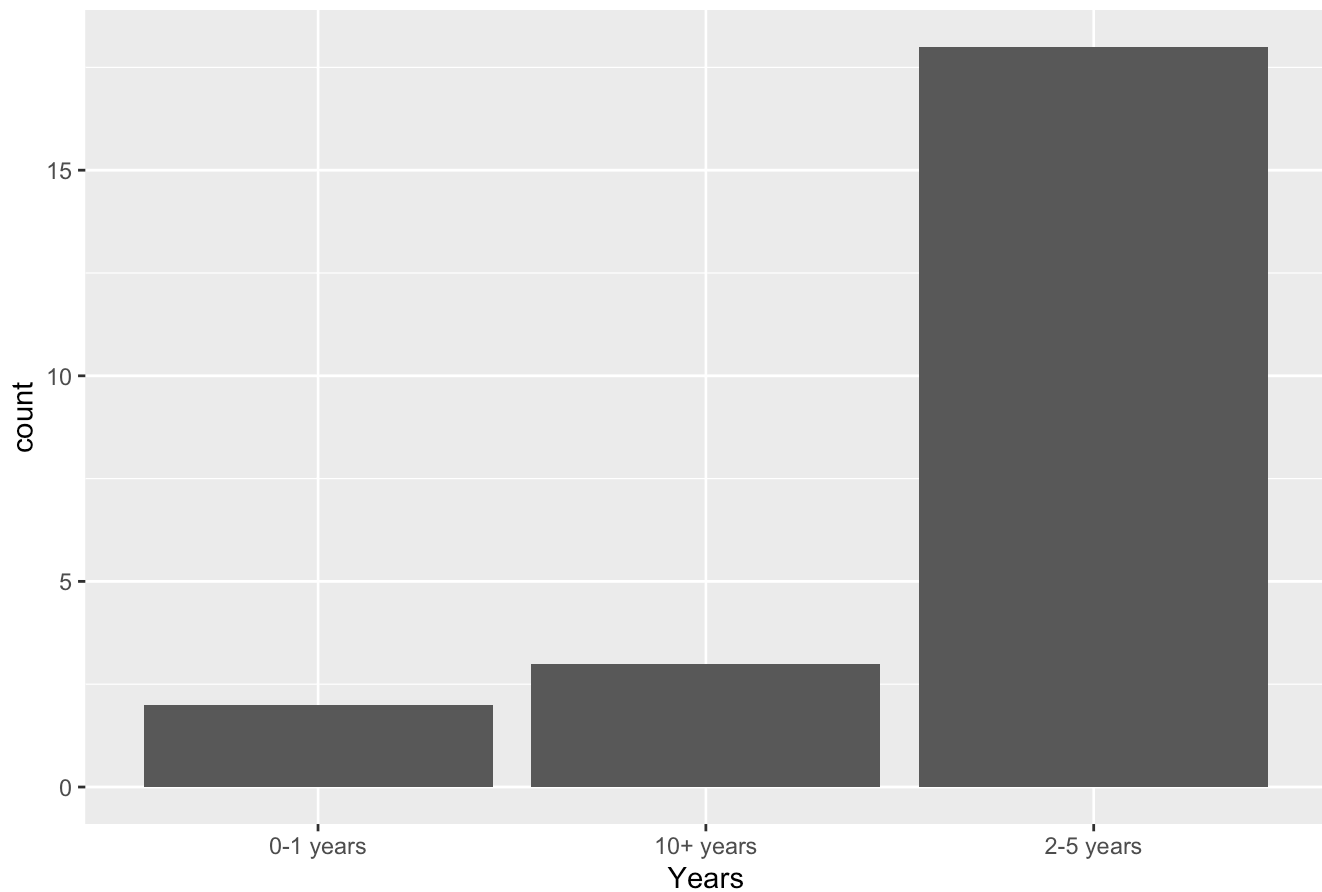
```
0-1 years 10+ years 2-5 years
      2         3         18
```

```
prop.table(table(Bozeman$years_live_boz))
```

```
0-1 years 10+ years 2-5 years
0.08695652 0.13043478 0.78260870
```

```
Bozeman |>
  ggplot(aes(x=years_live_boz)) +
  geom_bar() +
  labs(title = "Years Lived in Bozeman", x="Years")
```

Years Lived in Bozeman



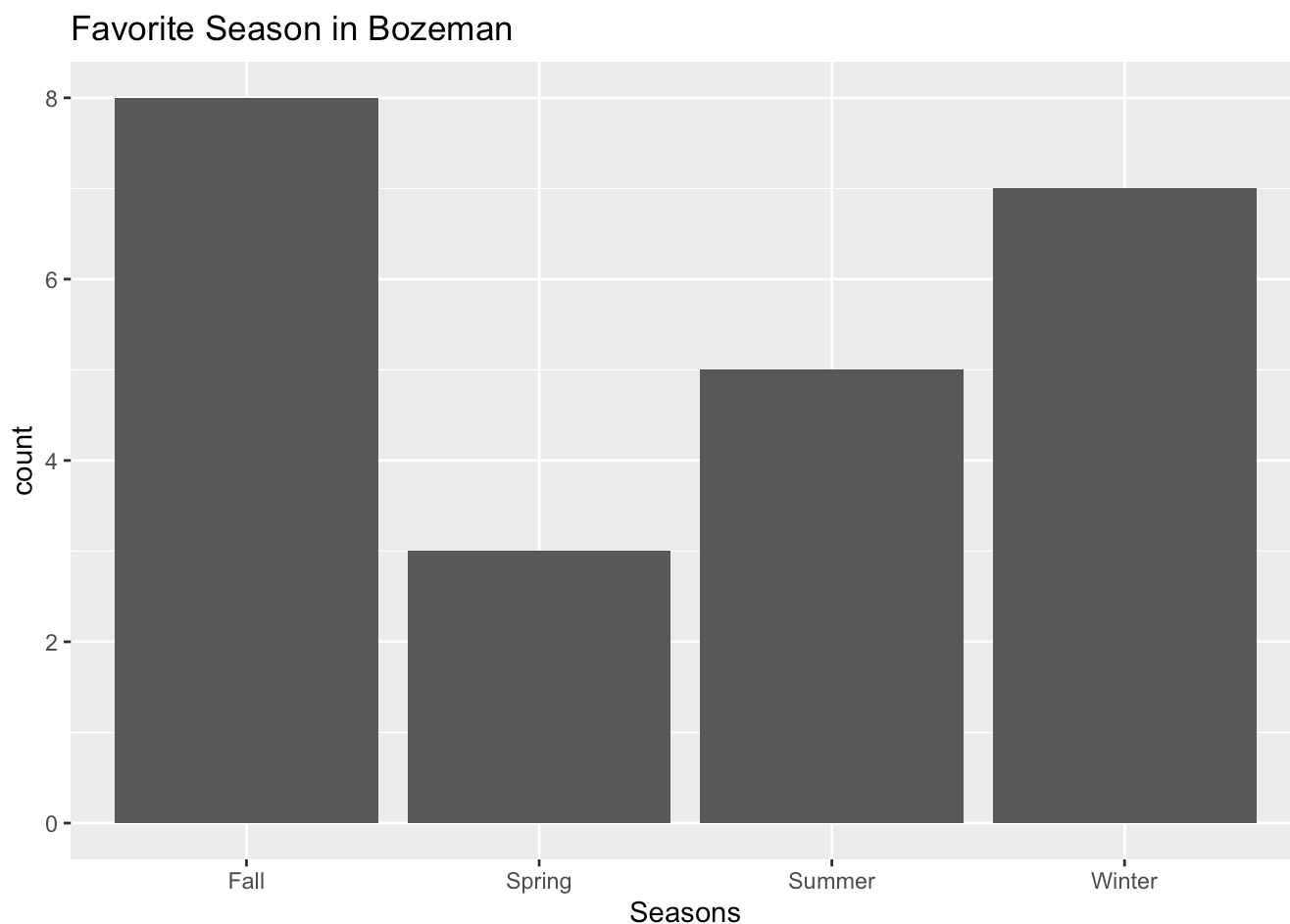
```
table(Bozeman$fav_season_boz)
```

```
Fall Spring Summer Winter
 8      3      5      7
```

```
prop.table(table(Bozeman$fav_season_boz))
```

```
Fall      Spring      Summer      Winter
0.3478261 0.1304348 0.2173913 0.3043478
```

```
Bozeman |>
  ggplot(aes(x=fav_season_boz)) +
  geom_bar() +
  labs(title = "Favorite Season in Bozeman", x = "Seasons")
```



- Numeric variables are `home_to_campus_miles_boz` and `msu_credits`.

```
colnames(Bozeman)
```

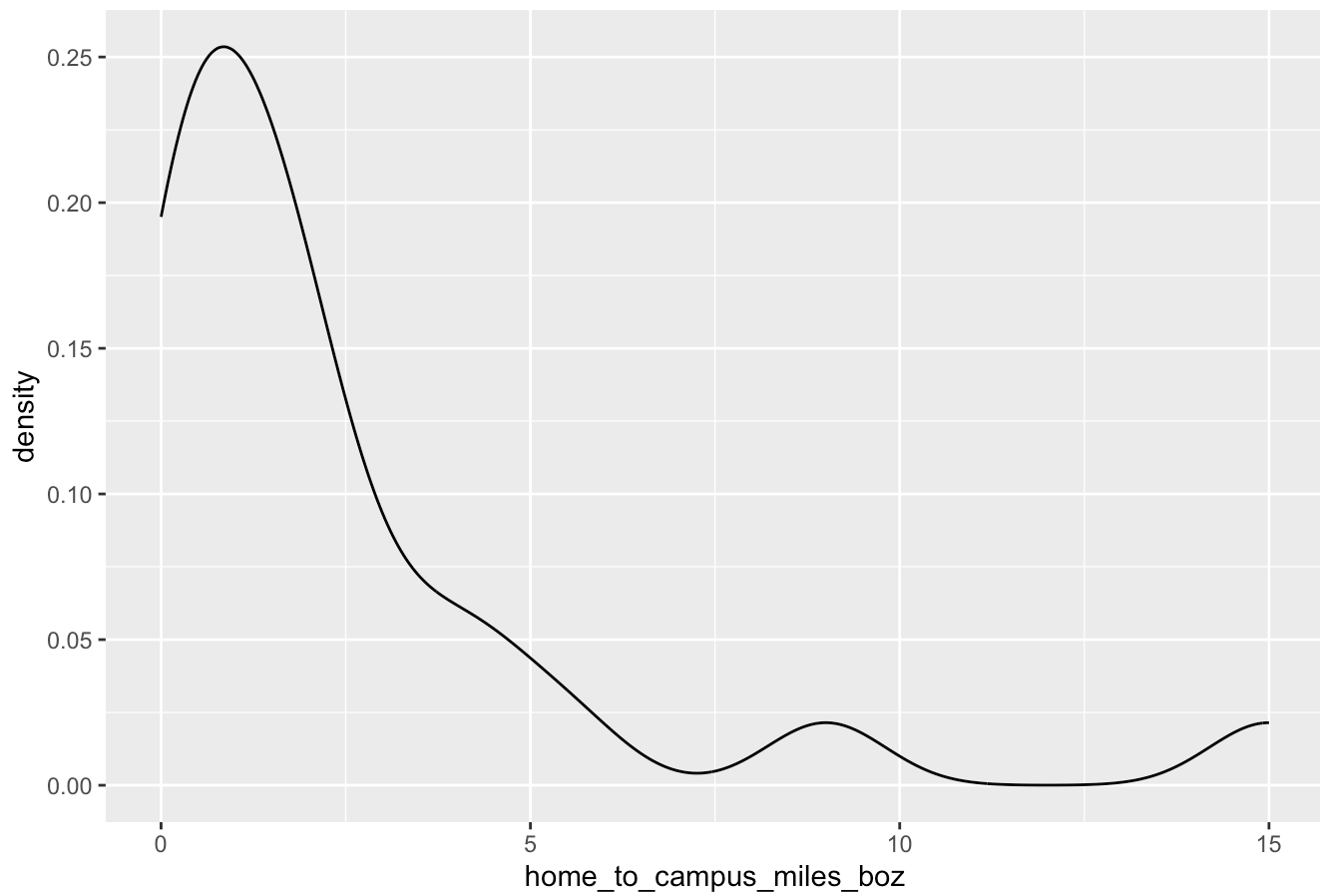
```
[1] "timestamp"           "years_live_boz"
[3] "for_fun_boz"         "part_boz_live"
[5] "fav_thing_boz_why"   "fav_season_boz"
[7] "fav_least_rest_boz" "home_to_campus_miles_boz"
[9] "msu_credits"
```

```
Bozeman$home_to_campus_miles_boz <-  
  as.numeric(gsub("[^0-9.]", "", Bozeman$home_to_campus_miles_boz))
```

```
Bozeman$msu_credits <-  
  as.numeric(gsub("[^0-9.]", "", Bozeman$msu_credits))
```

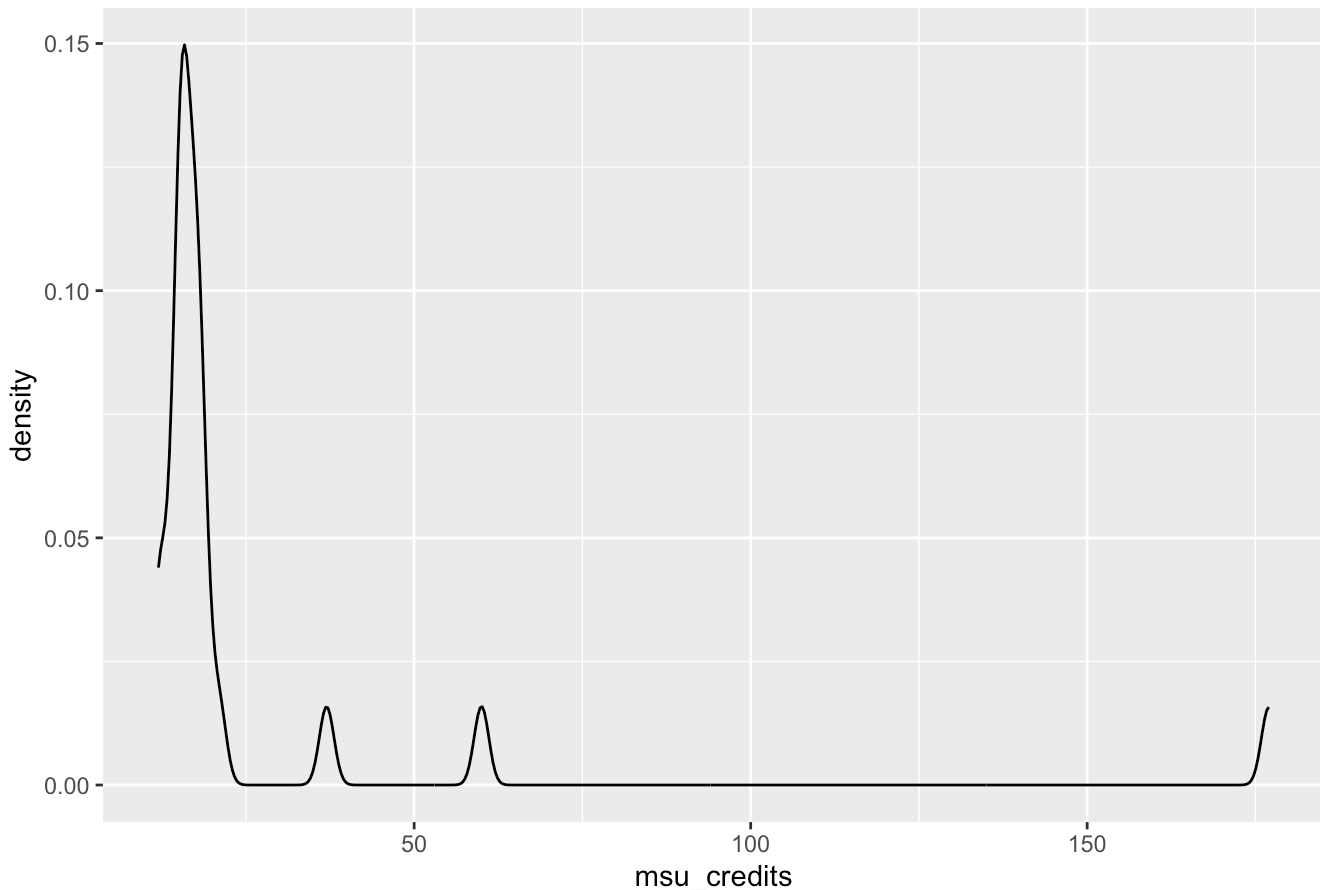
```
Bozeman |>  
  ggplot(aes(x= home_to_campus_miles_boz)) +  
  geom_density() +  
  labs(title = "Distance from Home to Campus")
```

Distance from Home to Campus



```
Bozeman |>  
  ggplot(aes(x= msu_credits)) +  
  geom_density() +  
  labs(title = "Credits Currently Taken at MSU")
```

Credits Currently Taken at MSU



- The plot showing the distribution of the distance in miles from ones home to campus has a fairly normal shaped but skewed distribution, it all has only one mode, it doesn't appear to have any extreme outliers, and data is spread out around an mean of 1 mile.
- The plot showing the distribution of ones msu credits is severely right skewed meaning there may be one or two extreme values in the data causing it to be pulled to the right. There does appear to be multiple modes and the data peaks around a mean of 15 credits.
- The columns with missing data are `fav_least_rest_boz` and `part_boz_live`. The pattern I am seeing in the NA or missing values, is that those taking the questionnaire either answered one or the other of the questions but not both and not neither.
- It's possible that there are outliers as both of the distributions I plotted are skewed but without doing th IQR test, we can't know for sure.

Manipulate the Data

```
Bozeman$credits <- ifelse(Bozeman$msu_credits > 15, "TRUE", "FALSE")
```

- The above uses the `ifelse()` function to display a new column called `credits` and this displays a True if the column meets the requirement of >15 credits and False if it does not.

```
Bozeman <- Bozeman |>
mutate(for_fun_bozo = case_when(
  for_fun_boz %in% c("Hiking", "hike", "Hike") ~ "Hike"
))
```

```
library(tidyverse)
```

```
Bozeman <- Bozeman |>
mutate(distance_travel_campus = case_when(
  home_to_campus_miles_boz < 1 ~ "Close",
  home_to_campus_miles_boz >= 1 & home_to_campus_miles_boz < 5 ~ "Medium",
  home_to_campus_miles_boz >= 5 ~ "Far"
))
```

```
Bozeman
```

	timestamp	years_live_boz	for_fun_boz
1	2026/02/26 12:46:11 PM MST	2-5 years	Hike
2	2026/02/26 12:47:24 PM MST	2-5 years	Go downtown
3	2026/02/26 12:47:31 PM MST	2-5 years	Hike
4	2026/02/26 12:47:41 PM MST	2-5 years	Climb rocks and ride bikes
5	2026/02/26 12:47:51 PM MST	2-5 years	Climb rock
6	2026/02/26 12:48:16 PM MST	2-5 years	get a cute little coffee
7	2026/02/26 12:48:23 PM MST	2-5 years	Ski
8	2026/02/26 12:48:30 PM MST	2-5 years	walk
9	2026/02/26 12:49:02 PM MST	2-5 years	Walk around my neighborhood.
10	2026/02/26 12:49:03 PM MST	2-5 years	Go on bike rides
11	2026/02/26 12:49:26 PM MST	2-5 years	I like to hike hyalite canyon
12	2026/02/26 12:49:52 PM MST	10+ years	Walk
13	2026/02/26 12:50:16 PM MST	10+ years	hike
14	2026/02/26 12:51:01 PM MST	10+ years	Camp
15	2026/02/26 12:52:03 PM MST	2-5 years	Ski
16	2026/02/26 12:52:06 PM MST	0-1 years	Hiking
17	2026/02/26 12:52:06 PM MST	2-5 years	Go skiing
18	2026/02/26 12:52:50 PM MST	2-5 years	goon
19	2026/02/26 12:53:16 PM MST	0-1 years	Tanning
20	2026/02/26 12:53:52 PM MST	2-5 years	Trivia at the Molly Brown
21	2026/02/26 12:53:55 PM MST	2-5 years	Hike
22	2026/02/26 12:54:07 PM MST	2-5 years	Hike
23	2026/02/26 12:54:29 PM MST	2-5 years	Ski

	part_boz_live
1	Near Campus
2	North of main
3	South/Close to campus
4	Central
5	North East
6	the near campus part?
7	
8	south Bozeman

9
10
11
12
13 close to campus
14 19th
15 Central
16 On Campus
17 Babcock
18 south
19 Belgrade
20 off babcock
21
22
23 Near campus

fav_thing_boz_why

1
All of the outdoors activities because who doesn't want to be outside
2
The community and how many different groups you can be a part of
3 My favorite thing about Bozeman is that it is so similar to Missoula. I'm from
Missoula, and I love it there but it's nice to get a little ways away. Bozeman is that,
but still very similar to the hometown I love.
4
The proximity to recreational Opportunities
5
The people! many outdoorsy people like me
6
The people that I've met because they're the best friends I've ever had.
7
The weather
8
I like how close it is to various national parks and forests and how close it is to
fossil lands.
9
The people – I love everyone I have met here.
10 The community and
all the things do to. It is so awesome just being able to go up to Bridger and always see
someone that you know in the liftlines.
11
I like the downtown area, its fun to walk around and take friends to
12
The mountains
13
winter because it scares the old people away
14
Close to mountains cause I like them
15
Mountains
16

The people and the location because its awesome

17

How close it is to the mountains

18

the weather

19

Target and Tj max

20

its perty

21

MSU

22

Close to Yellowstone, because I like Yellowstone

23

Cougars cause I wouldn't have it any other way

fav_season_boz

1 Summer

2 Winter

3 Fall

4 Winter

5 Winter

6 Fall

7 Fall

8 Fall

9 Fall

10 Winter

11 Fall

12 Summer

13 Winter

14 Spring

15 Spring

16 Fall

17 Summer

18 Spring

19 Winter

20 Fall

21 Summer

22 Winter

23 Summer

fav_least_rest_boz

1

2

3

4

least favorite is probably subway.

5

Western cafe, and shawarma bus

6

7

Mr Burritos and least is Daves Sushi

Favorite is Ramires,

Favorite is

8
 9 Favorite -
 Blackbird. Least favorite - McDonalds
 10 My favorite resturant in bozeman J.W. Heist, and my
 least favorite is probabaly Wendy's.
 11 My favorite restaurant is
 5 on black, least favorite is arby's
 12
 Mcdonalds
 13
 14
 15 Favorite restaurant is Hooked
 Sushi, Least is the chipotle on main
 16 Favorite is that one poke
 bowl place, least favorite is taco bell
 17
 18
 fav- spits least fav- town pump
 19 Favorite is seven sushi bc it's really good. My least favorite is the sushi
 restaurant off of main b/c its gross.
 20 Favorite - Los Jarachos,
 least favorite - Miller dinning hall
 21 Cholms and I dont like Little Ceasers because I went in ten minutes before close one
 time and they wouldn't give me pizza.
 22
 Favorite: Ramirez tacos, Least favorite:

23

	home_to_campus_miles_boz	msu_credits	credits	for_fun_bozo
1	2.00	21	TRUE	Hike
2	1.00	13	FALSE	<NA>
3	0.50	16	TRUE	Hike
4	0.40	16	TRUE	<NA>
5	3.00	12	FALSE	<NA>
6	0.50	37	TRUE	<NA>
7	1.50	18	TRUE	<NA>
8	0.00	18	TRUE	<NA>
9	5.50	12	FALSE	<NA>
10	2.00	16	TRUE	<NA>
11	4.30	177	TRUE	<NA>
12	9.00	15	FALSE	<NA>
13	0.50	15	FALSE	Hike
14	1.50	16	TRUE	<NA>
15	2.50	18	TRUE	<NA>
16	0.10	15	FALSE	Hike
17	1.50	15	FALSE	<NA>
18	0.14	15	FALSE	<NA>
19	15.00	60	TRUE	<NA>
20	4.00	19	TRUE	<NA>
21	1.50	17	TRUE	Hike
22	0.00	17	TRUE	Hike
23	1.00	18	TRUE	<NA>

	distance_travel_campus
1	Medium
2	Medium
3	Close
4	Close
5	Medium
6	Close
7	Medium
8	Close
9	Far
10	Medium
11	Medium
12	Far
13	Close
14	Medium
15	Medium
16	Close
17	Medium
18	Close
19	Far
20	Medium
21	Medium
22	Close
23	Medium